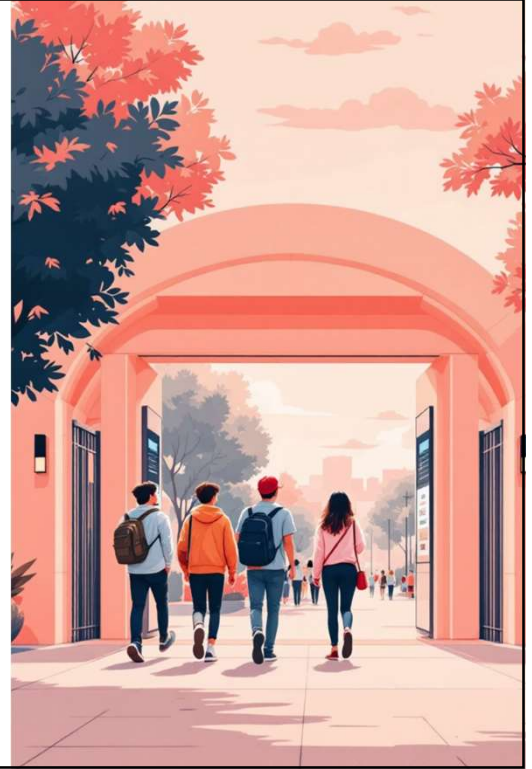


Gatepass Monitoring System

A digital solution to manage and track student movement on campus.



Overview & Team

This system streamlines gate pass requests, approvals, and verification, enhancing campus security and reducing manual effort.

Guide

Divya S Nair

Group Members

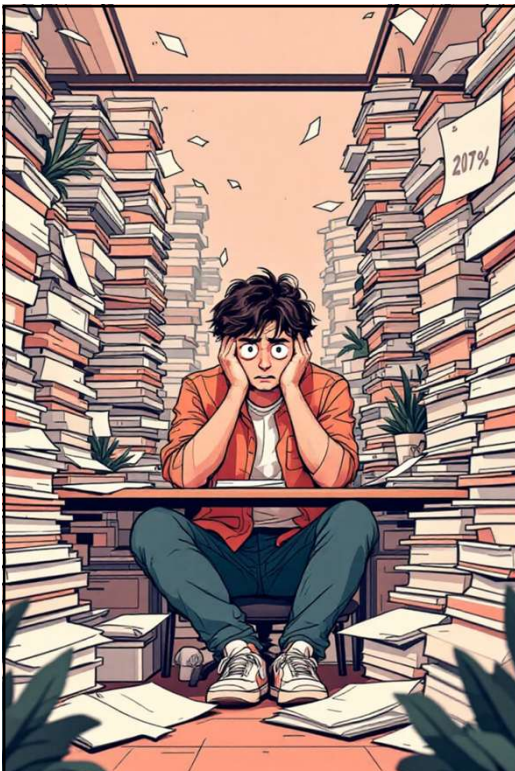
- Pavithra J J
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Problem Statement: Manual Inefficiencies

Time-Consuming

Manual registers and paper passes are inefficient.

Error Prone

High chances of errors, lost records, and data duplication.

Lack of Control

Difficulty tracking entry/exit and controlling unauthorized movements.

No Real-time Data

Absence of centralized system for monitoring and reporting.

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Project Roadmap: Key Phases

01

Project Planning

Define objectives, scope, and identify users (Student, Admin, Security, Parent).

02

Requirement Analysis

Gatepass request & approval, security verification, parent alerts.

03

Feasibility Study

Technical, economic, and operational feasibility assessment.

04

System Design

Architecture, Use case, DFD, and ER diagrams.

05

Technology Selection

Choose frontend, backend, database, and communication services.

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Roadmap: Implementation Details

01

Database Design

User, parent details, gatepass, and message logs.

02

User Module

Login and gatepass request submission.

03

Admin Module

Approve/reject gatepass, trigger notifications.

04

Gatepass Generation

Unique ID creation and status updates.

05

Security Verification

Gate verification and log entry/exit times.

06

Parent Notification

SMS/email alerts and message status storage.

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Literature Survey: Manual vs. Automated

Manual Gatepass System

Paper-based registers for student sign-outs and faculty signatures.

Advantages

- Simple and inexpensive
- No technical setup needed

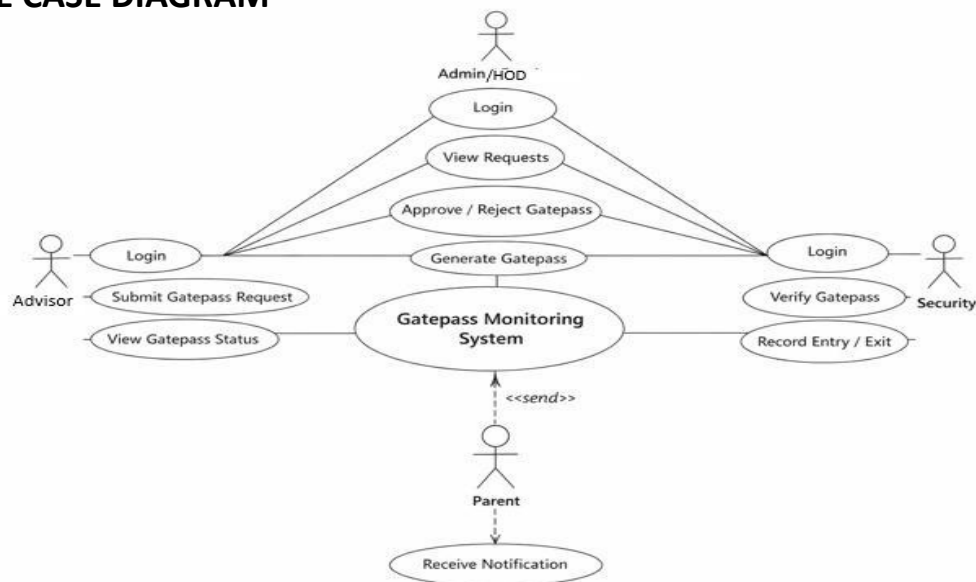
Disadvantages

- Time-consuming and inefficient
- Prone to errors or lost records
- No real-time tracking

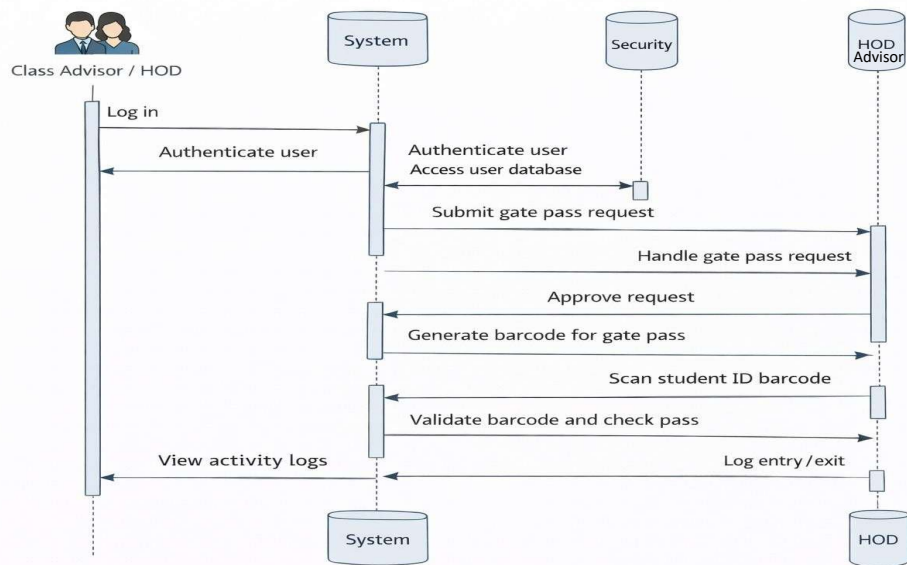


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USE CASE DIAGRAM

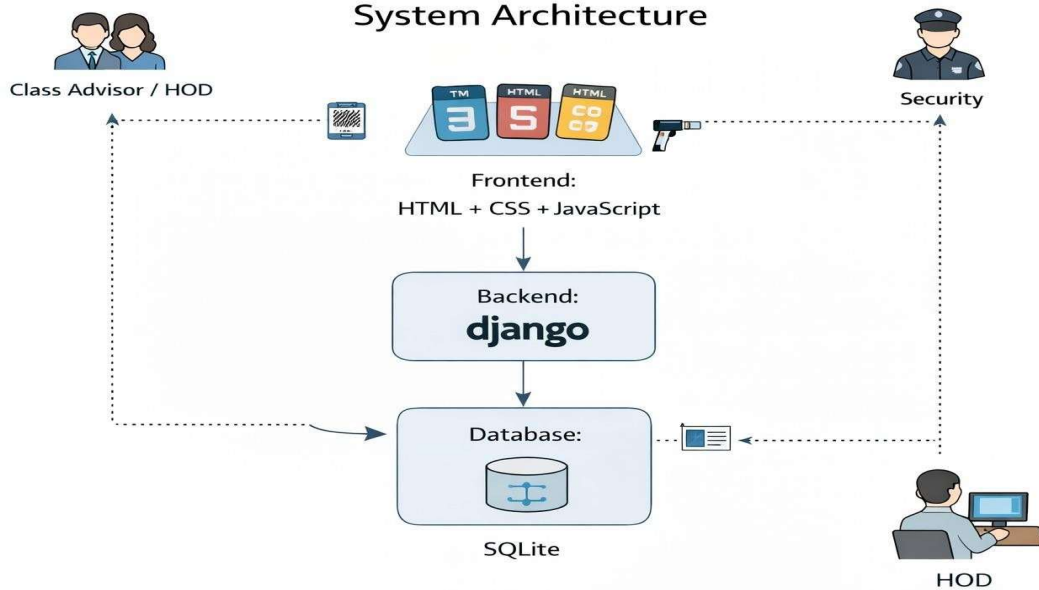


Sequence Diagram



SystemArchitecture

System Architecture



System Implementation: Technology Stack



User Roles

Admin (Class Advisor/HOD), Security, HOD (monitoring).



Frontend

HTML, CSS, JavaScript for login and barcode scanning.



Backend

Django for authentication, gate pass generation, and approval.



Database

SQLite for user, gate pass, barcode, and log data.

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Implementation: Process Flow

1

Gate Pass Generation

Advisor/HOD scans ID, system generates pass, parent notified.

2

Gate Pass Verification

Security scans ID, system checks validity, allows/denies exit/entry.

3

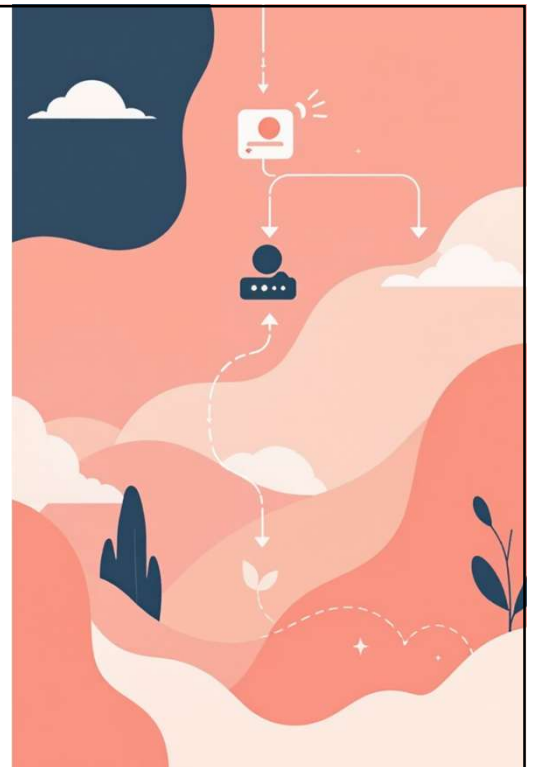
Entry/Exit Logging

Automatic recording of entry, exit, and re-entry times.

4

Monitoring & Reports

HOD views passes, current exits, re-entries, and generates reports.



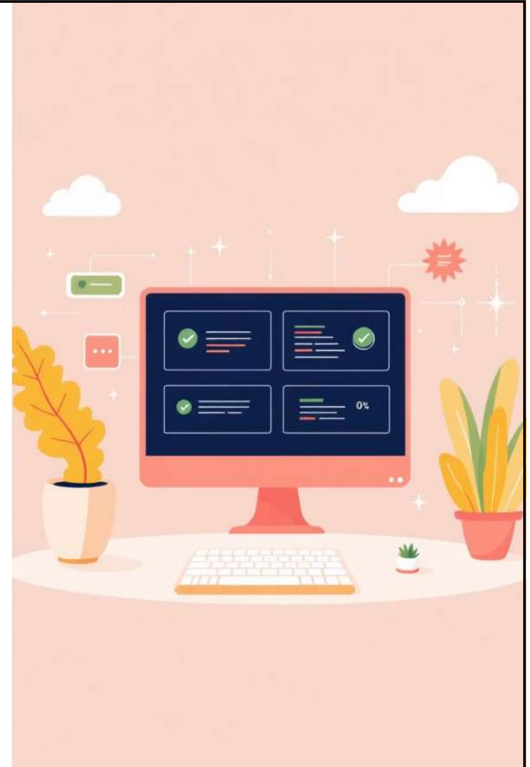
Results & Benefits

Key Results

- Student requests submitted online.
- Admin approval/rejection efficient.
- Unique gatepasses generated.
- Security verification accurate.
- Parents receive automatic notifications.

System Benefits

- Eliminates manual paperwork.
- Improves campus security.
- Real-time monitoring.
- Accurate tracking.
- Role-based access control.



Conclusion: Enhanced Campus Management

The Gatepass Monitoring System successfully digitizes the entire process, ensuring efficiency, security, and transparency. It enhances student safety through real-time notifications to parents and improves overall campus workflow, accountability, and communication.